Qualitative Data Analysis

Dirk Riehle, FAU Erlangen

NYT B03

Course Content and Structure

Overview

1. What is science?
2. Scientific research
3. **Qualitative data analysis**

Theory building

4. Theory building
5. Systematic reviews
6. Qualitative surveys
7. Action research
8. Case study research

Theory validation

9. Theory validation
10. Survey research
11. Controlled experiments

Comprehensive

12. Design science

Academia

13. Academic writing
14. Academic publishing

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Agenda

1. QDA in theory building
2. QDA artifacts
3. QDA methods
4. Straussian coding
5. Quality assurance
6. Iterative process
7. Theory presentation

1. QDA in Theory Building

Qualitative Data Analysis (QDA) [1]

Qualitative data analysis is a research method

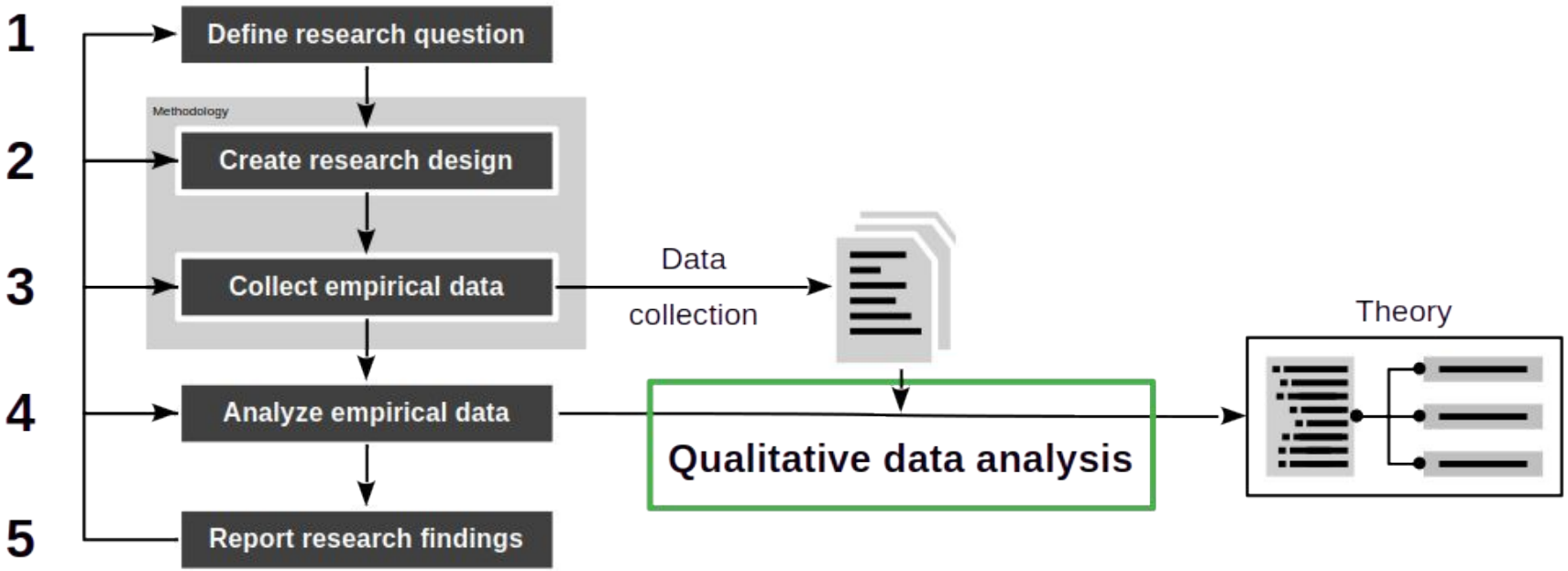
- For deriving meaning from qualitative empirical data
- In response to one or more research questions

Typically used in theory building and development

- Inductive use → theory building
- Deductive use → theory evaluation

[1] Also qualitative content analysis (German: qualitative Inhaltsanalyse)

Position and Purpose of QDA in Theory Building



QDA in Theory Building Methodologies

Most QDA methods can be plugged into theory building methodologies

- Systematic reviews
- Qualitative surveys
- Action research
- Case study research

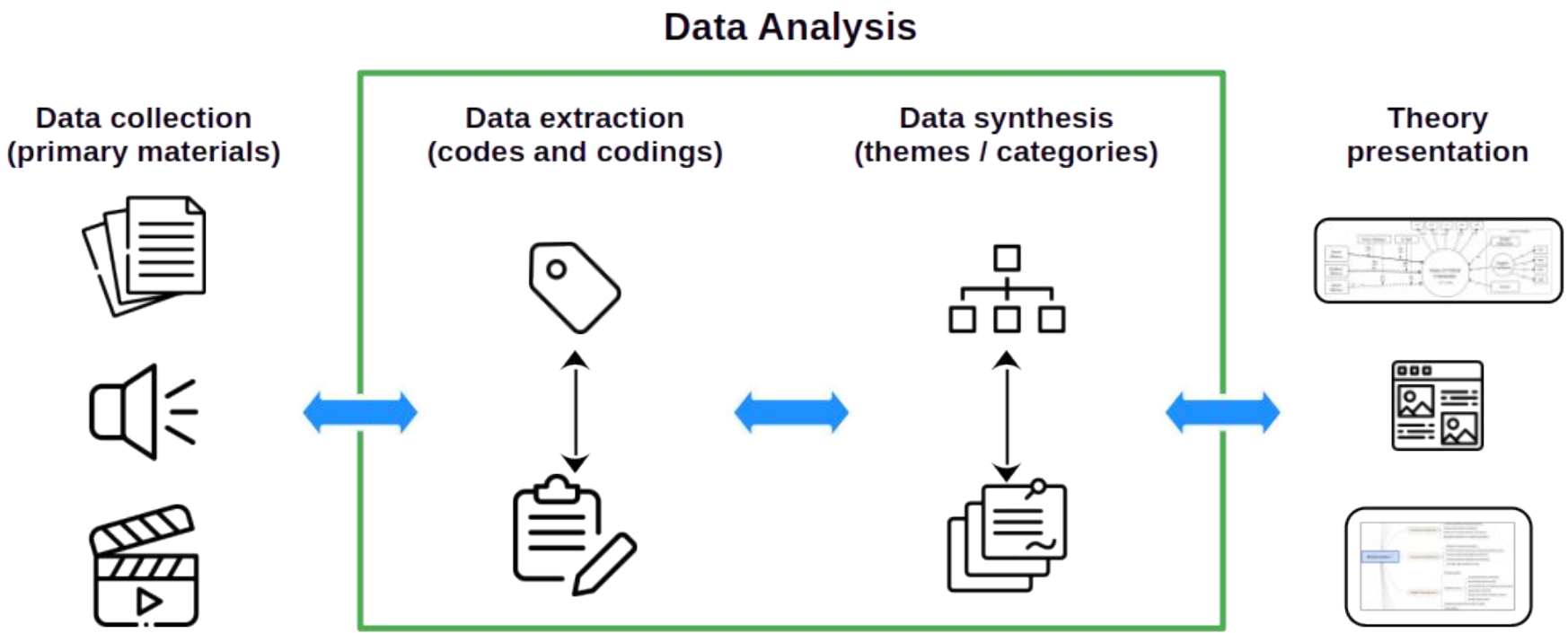
Properties of QDA

The qualitative data analysis process is

- Inductive (→ theory building from data)
- Structured (→ not arbitrary / random)
- Quantitative (→ measurable quality)

2. QDA Artifacts

Artifacts and Activities in QDA



Codes and Codings

A code (also label, annotation, etc.) is

- A short text (name) for an empirical observation

An open code is a code that

- Originates from the primary materials through a coding

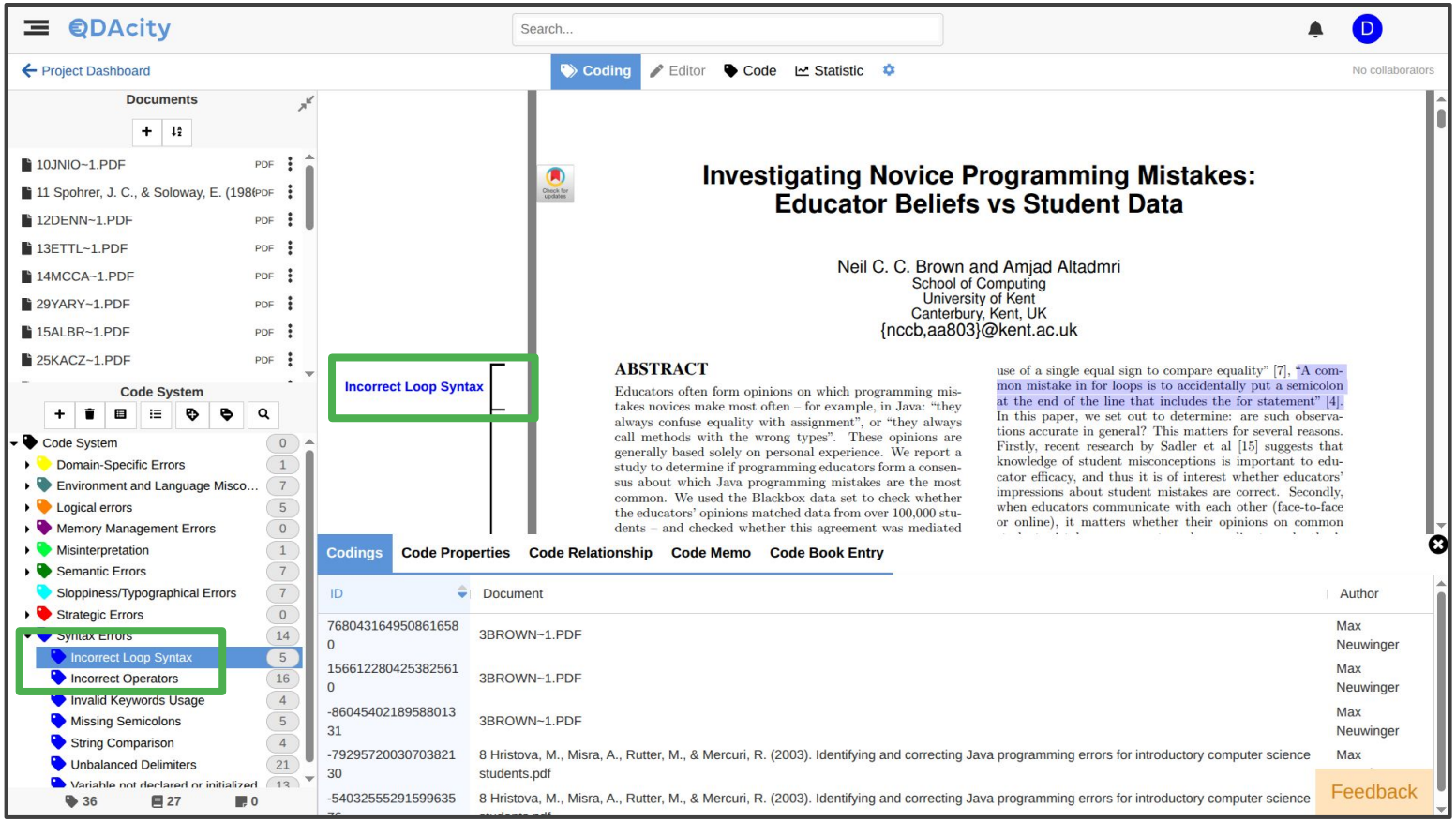
A coding is

- The relationship between an open code and its origin in the data

A theme or category is a code that

- Hierarchically abstracts from other codes in a code system

Example Code and Codings



Code Definition

A code definition should include

- A short (one sentence) definition
- A description of
  - When to use the code and
  - When not to use the code

Memo

A memo is

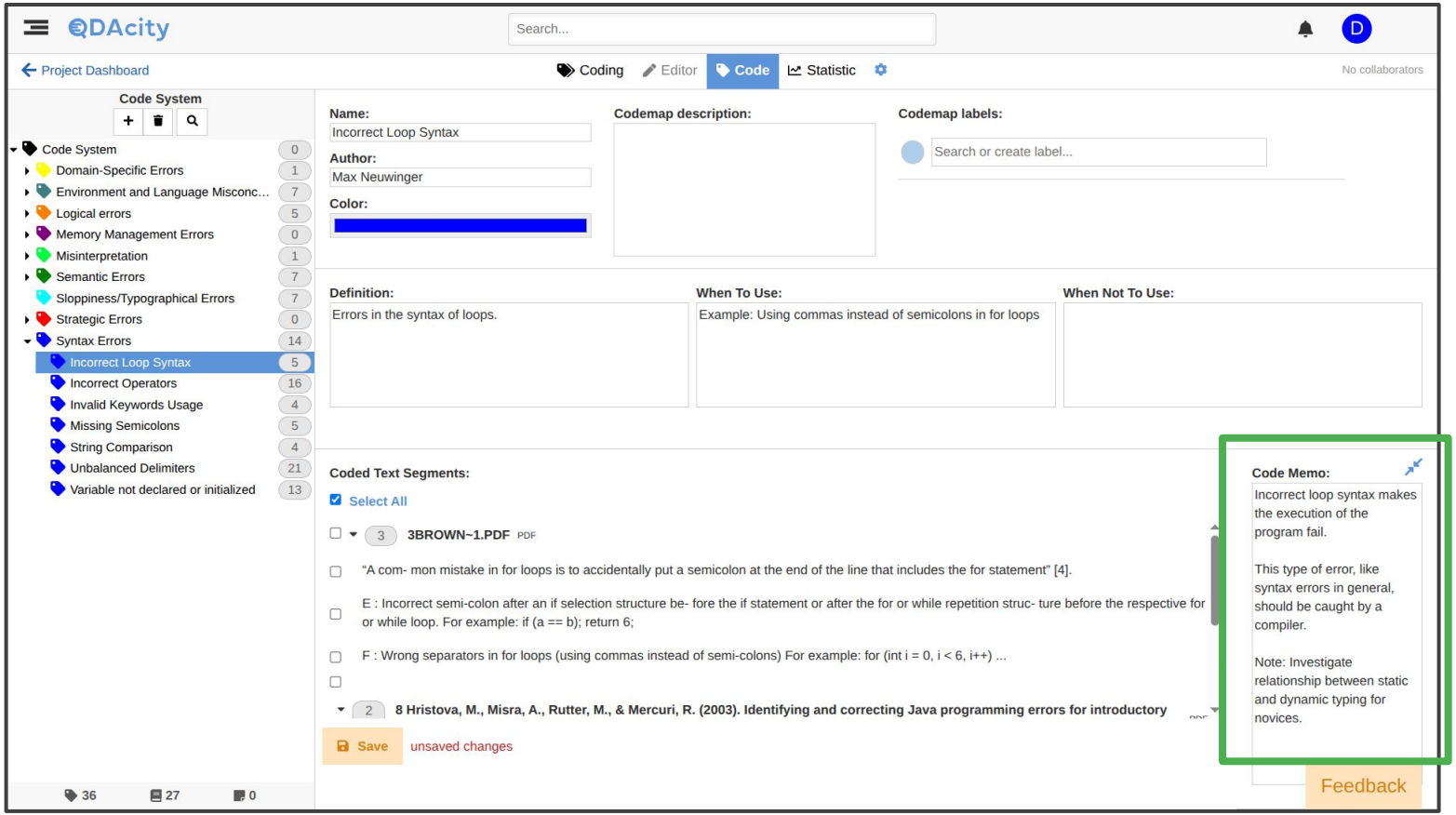
- A written reflection on a code

A memo may include

- Research insights and observations
- Work notes for future investigations

The code itself should be defined independently of the memo

Example Code and Memo



Code System

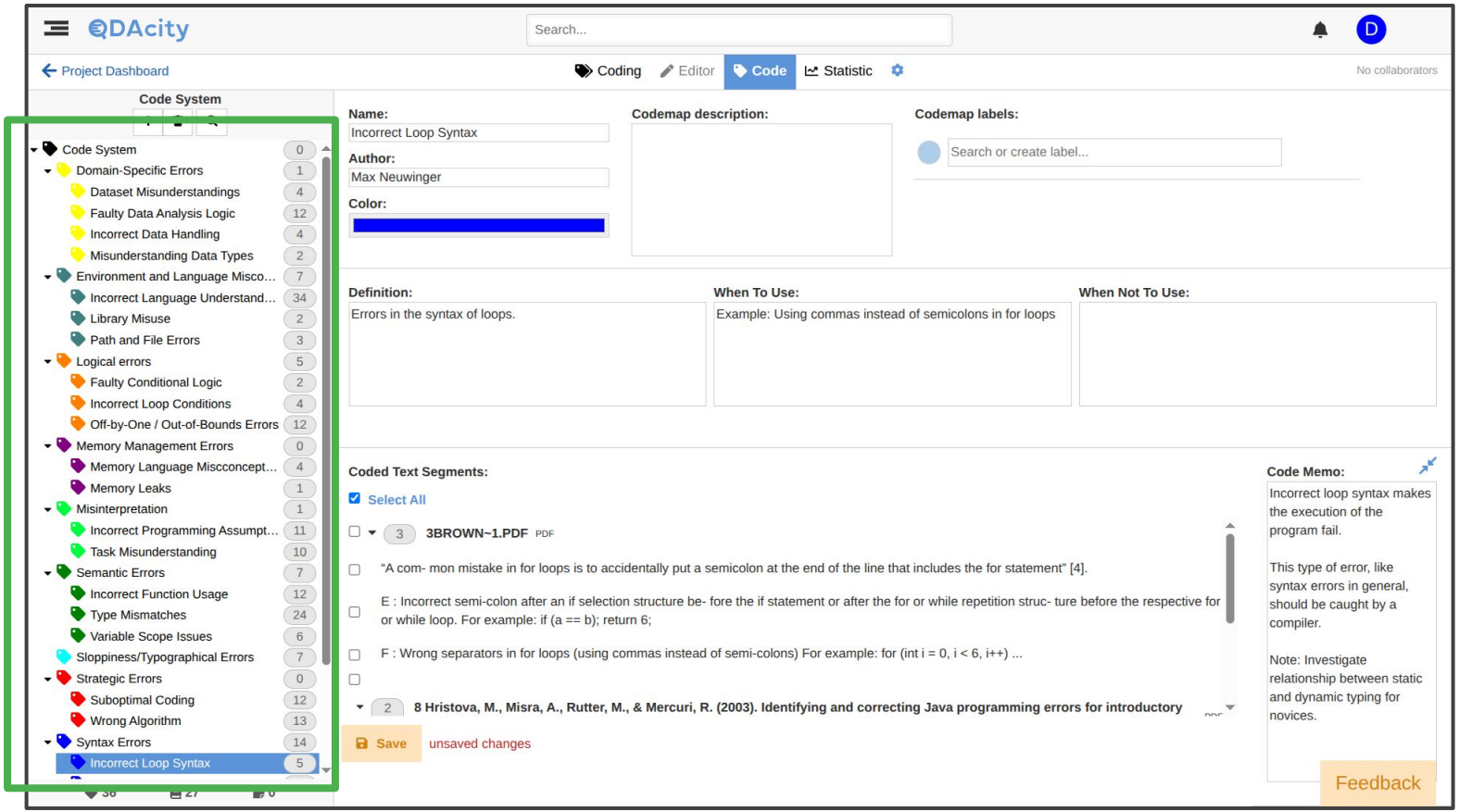
A code system is

- A hierarchical structure of codes intended
- To capture the relationships between the codes

A code system is an expression of the theory under development

- Codes represent concepts and constructs
- Relationships represent hypotheses

Example Code System



The Semantics of Hierarchy

In principle, anything goes, but …

Common patterns of hierarchical organization are

- A slotting system to group similar open codes (cf. card sorting)
- A grouping system for underlying themes (cf. thematic analysis)
- A structuring mechanism for theoretical constructs / abstractions

Computer science students may use common modeling techniques like

- Association, aggregation (part-whole), generalization/specialization

No tool (yet) allows for the formal capture of such (meta) relationships

Code Book

A code book documents the code system

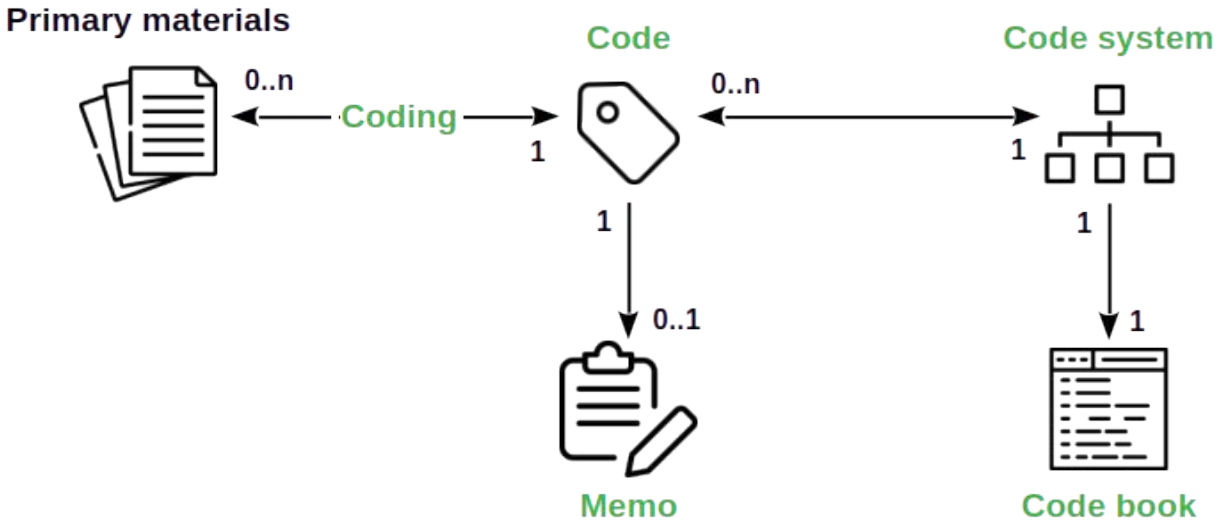
For each code in the code system, the code book lists

- The definition of the code
- Instructions for when to use the code
- Possible disambiguations towards other codes
- An exemplary piece of data coded with the code

Beyond general documentation, a code book

- Helps establish intercoder reliability of analysis

Main Artifacts in Overview



3. QDA Methods

Three Different QDA Methods

1. Card sorting [1]
2. Thematic analysis [2]
3. Straussian coding [3]

[1] See Hudson (2013): Card sorting.
[2] See Braun & Clarke (2012): Thematic analysis.
[3] See Corbin & Strauss (2008): Qualitative research, 3rd edition.

Card Sorting [1]

Analyze data

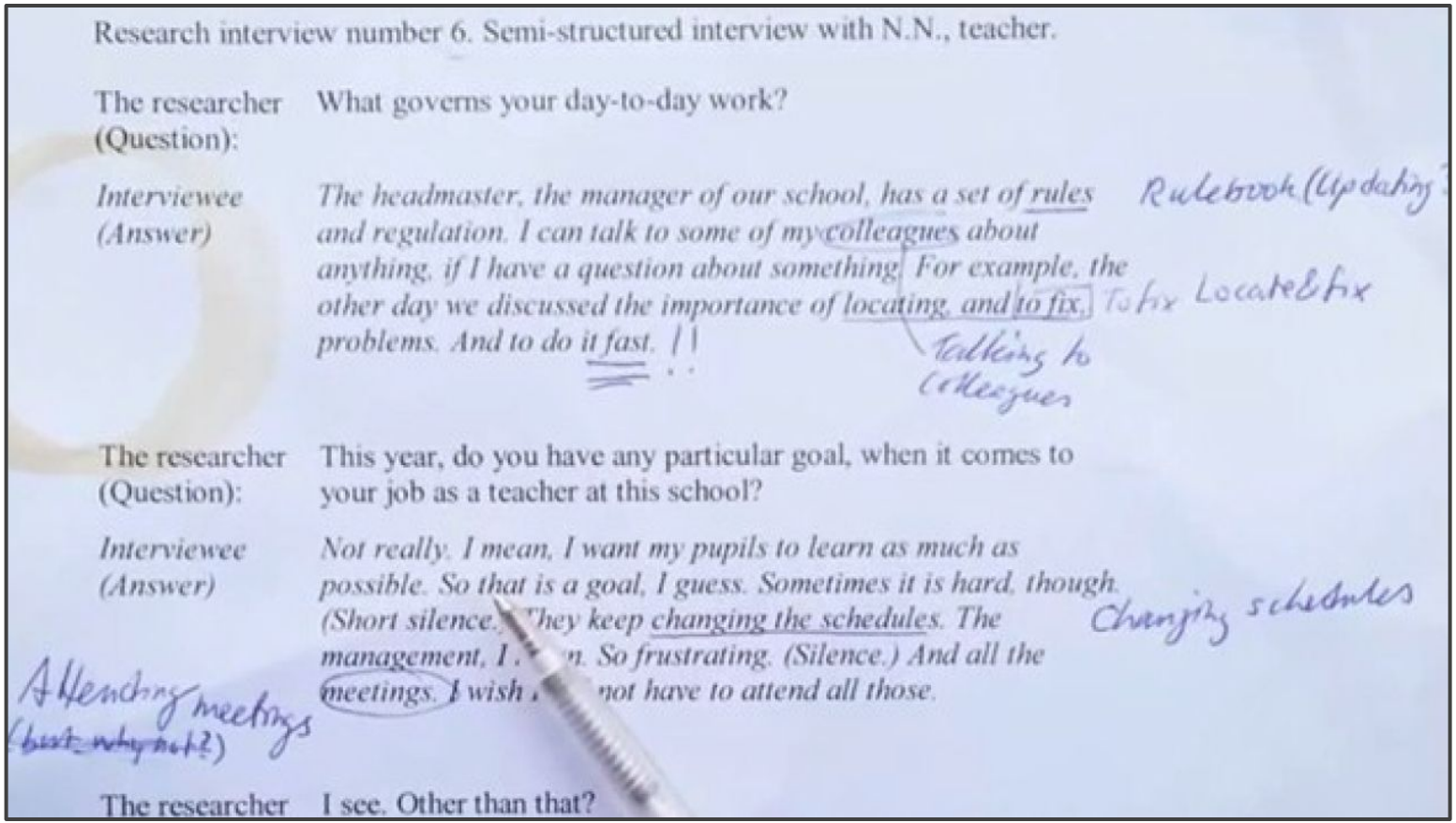
- Extract data
  - Name item [2] and write on card
- Synthesize data
  - Group cards with similar items

Codify theory

- Name and summarize groups

[1] See Hudson (2013): Card sorting.
[2] Item, topic, theme, artifact, … depends on context and is relative to research question

Coding, Then



Research interview number 6. Semi-structured interview with N.N., teacher.

The researcher (Question): What governs your day-to-day work?

Interviewee (Answer): *The headmaster, the manager of our school, has a set of rules and regulation. I can talk to some of my colleagues about anything, if I have a question about something. For example, the other day we discussed the importance of locating, and to fix, problems. And to do it fast. ||*

Rulebook (Updating?

To fix Locate & fix

Talking to colleagues

The researcher (Question): This year, do you have any particular goal, when it comes to your job as a teacher at this school?

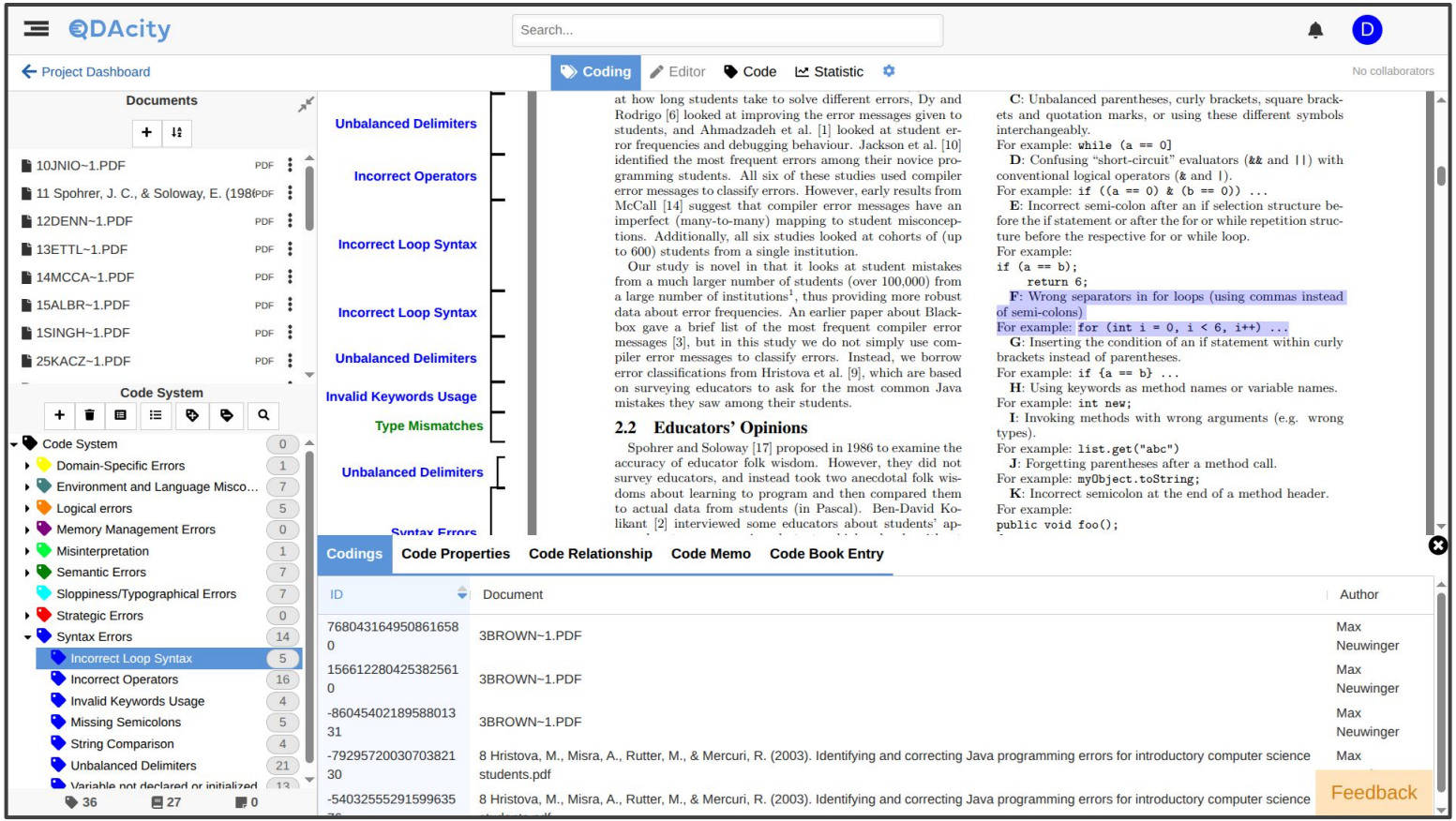
Interviewee (Answer): *Not really. I mean, I want my pupils to learn as much as possible. So that is a goal, I guess. Sometimes it is hard, though. (Short silence.) They keep changing the schedules. The management, I ... So frustrating. (Silence.) And all the meetings. I wish ... not have to attend all those.*

Changing schedules

Attending meetings (but why not?)

The researcher: I see. Other than that?

And Now (With QDAcity)



Thematic Analysis [1]

Analyze data

- Extract data
  - Familiarize yourself with the data
  - Generate initial codes
- Synthesize data
  - Search for potential themes
  - Review potential themes
  - Define and name themes

Codify theory

- Produce the report

[1] See Braun & Clarke (2012): Thematic analysis.

Straussian Coding [1]

Analyze data

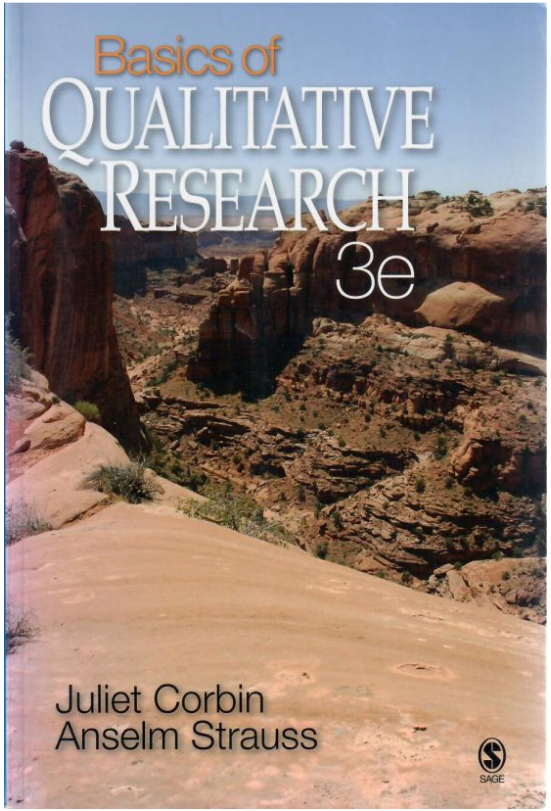
- Extract data
  - Perform open coding
- Synthesize data
  - Perform axial coding
  - Perform selective coding

Codify theory

- Write articles

[1] See Corbin & Strauss (2008): Qualitative research, 3rd edition.

Grounded Theory



Grounded theory is a research methodology for theory building like qualitative surveys, etc.

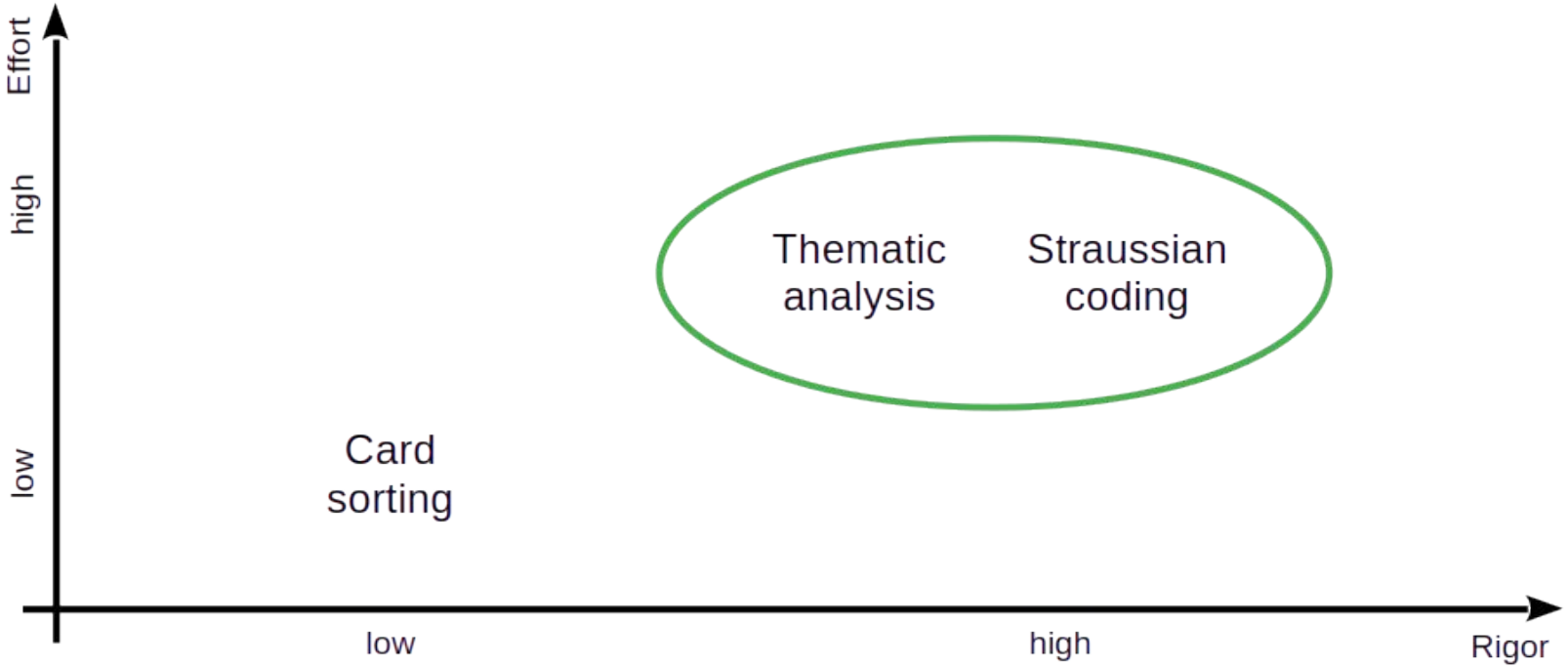
Straussian coding was lifted from Corbin & Strauss' version of grounded theory [1]

[1] See Corbin & Strauss (2008): Qualitative research, 3rd edition.

Artifacts and Activities in QDA Methods

		Card sorting	Thematic analysis	Straussian coding
Artifacts	Codes	Items	Codes	Open codes
	Categories	Item groups	Themes and subthemes	Concepts and categories
Activities	Extraction	Card writing	Initial coding	Open coding
	Synthesis	Grouping and summarizing	Theme naming and defining	Axial and selective coding

Rigor vs. Effort in QDA Methods



A Word of Caution

Despite my correlation and attempted homogenization

- Method authors and users can be extremely picky

So in a given research project, always use the specific method

4. Straussian Coding

Open Coding 1 / 2

An open code is a code that

- Represents an empirical observation
- Is conceptual, i.e. represents a concept
- Has a direct origin in the primary materials

Open coding is

- The creation of open codes from primary materials

An in-vivo code is an open code that

- Has a name taken directly from the primary materials

Open Coding 2 / 2

Carefully read the primary materials incrementally, repeatedly

Mark segments containing relevant information for your research question

Assign segment to existing open code or create a new one

Write memos for codes to elaborate on your observations

Constant Comparison

As you are coding, compare codes

- Are two codes the same?
- Are there subtle differences? Should the codes be split?

Then take action and refactor your code system

- Implies rereading and recoding

Axial Coding

Axial coding is

- The introduction of new codes representing concepts or constructs by
- Comparing existing codes for similarities and differences

Axial coding creates a hierarchy of codes, the code system

Codes should be MECE that is

- Mutually Exclusive and
- Completely Exhaustive

Concepts and Constructs

Codes represent concepts; examples

- Off-by-one error (open code)
- Syntax error (axial code)

Codes represent constructs; examples

- Product owner / software developer conflict (open code)
- Performance incentive (axial code)

Selective Coding

Selective coding is

- The focussing of top-level codes on the research question and
- The relationship-building of top-level codes to a single core code

According to Corbin & Strauss, the theory is the single core code

Resulting Theory

Code systems and associated artifacts represent the theory

There is more to the theory in a researcher’s mind

It needs research articles to document this

5. Quality Assurance

General Criteria for Qualitative Research

Quality criteria based on Guba & Lincoln [1]

- Credibility
- Transferability
- Dependability
- Confirmability

Typically realized through specific practices

[1] See Guba & Lincoln (1982): Naturalistic inquiry.

Example Research Practices

Two example research practices

- Member checking
- Peer debriefing

Fairly simple and easy

Member Checking

Member checking is a research practice in which the researcher

- Seeks feedback from data sources (participants, stakeholders) either
- During data gathering (active and reflective listening) or
- After analysis (asking for feedback and confirmation)

Peer Debriefing

Peer debriefing is a research practice in which the researcher

- Subjects their work to review and critique by a peer
- At multiple times during the current running project

The researcher presents and explains their work

The peer questions and critiques the work, looking for flaws

The researcher documents the session with a protocol

QDA-Specific Concerns

How do you know you are done?

⇒ Theory saturation

How do you know you are not winging it?

⇒ Intercoder agreement

Theory Saturation [1]

Theory saturation is a

- Stopping criterion for qualitative data analysis

Theory saturation is reached when

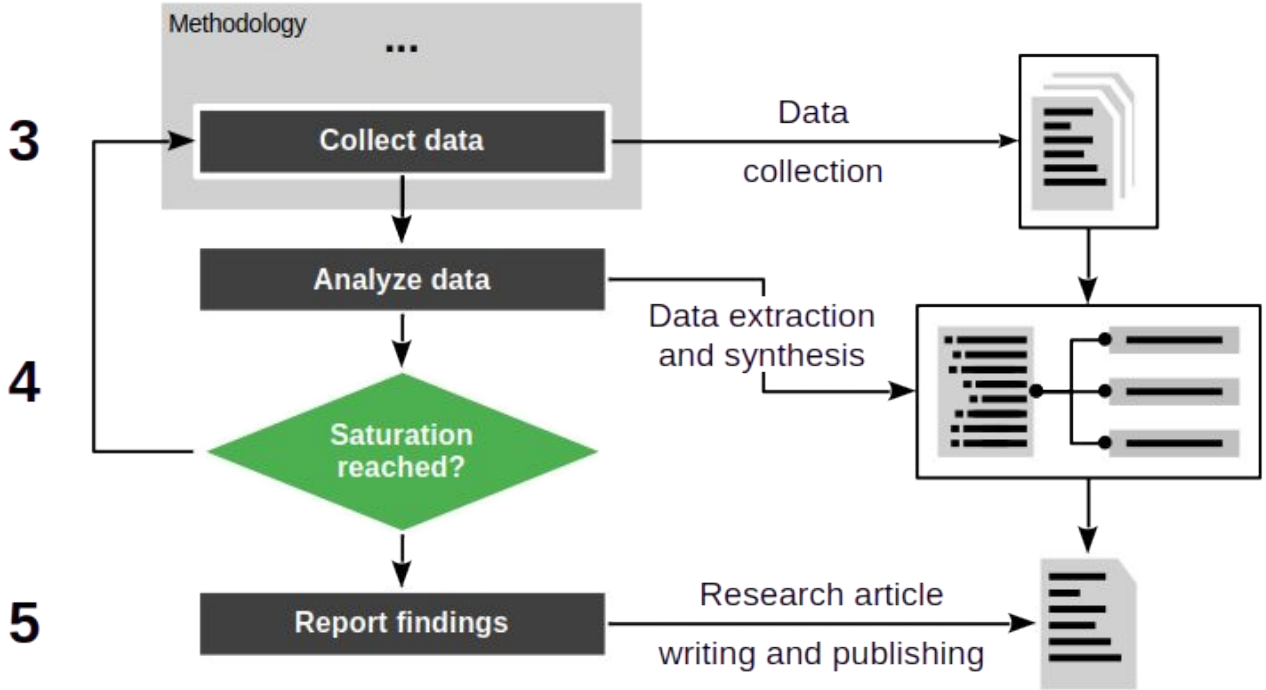
- Further data collection and analysis does not yield new insights

In measurable terms, the change rate of codes is zero

- New data yields new codings, but no new codes

[1] Originally theoretical saturation

Theory Saturation in Research Process



Measuring Theory Saturation

Structure analysis into iterations

- Use purposive sampling to seek out new data

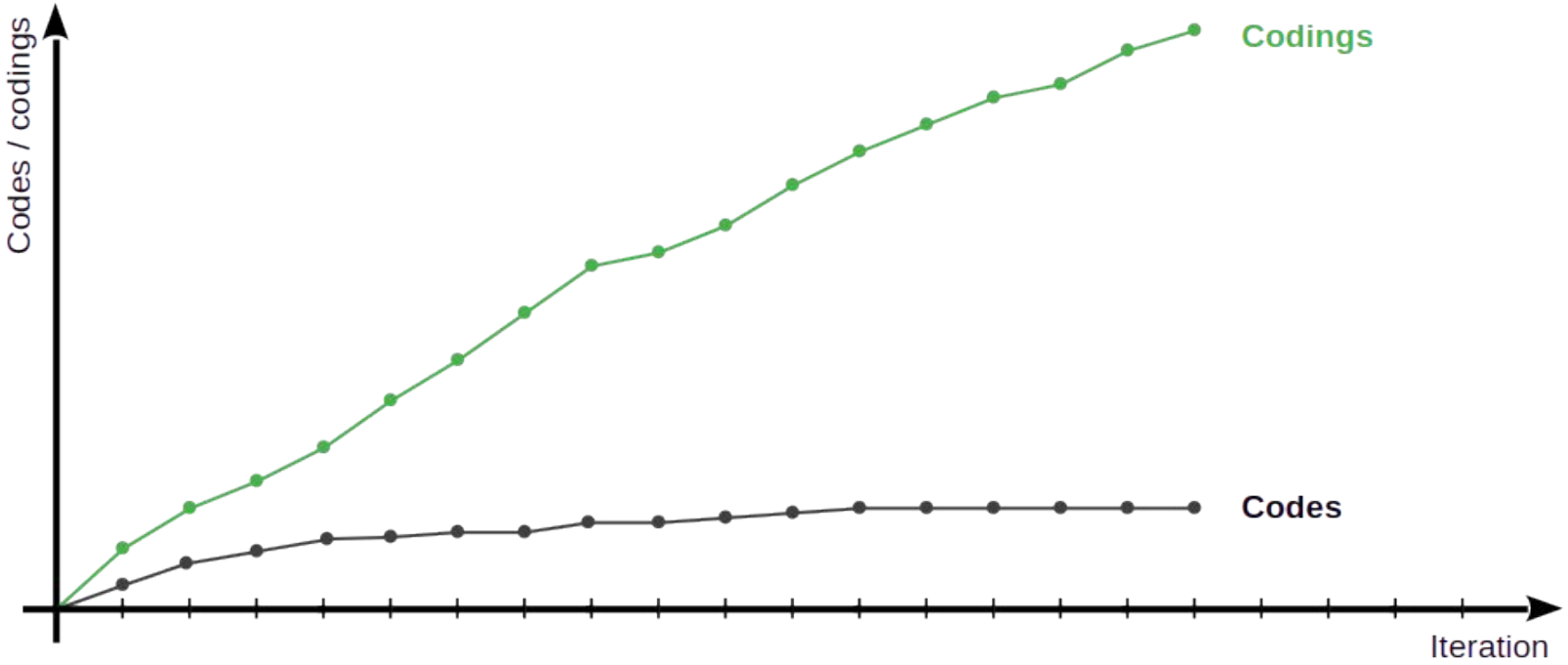
After finishing the analysis of new data

- Count the number of new codings and codes

No new codes in five iterations

- Theory saturation reached

Charting of Theory Saturation by Iteration



Intercoder Reliability

Intercoder reliability (also: interrater reliability) is

- The degree of agreement between
  - A primary (the original) coder and
  - One or more secondary coders

In other words: Is there intersubjective agreement?

- And to what extent?

Process of Measuring Intercoder Agreement

The original coder creates the code book

- The code book documents the code system

The original coder explains and discusses the code book with further coders

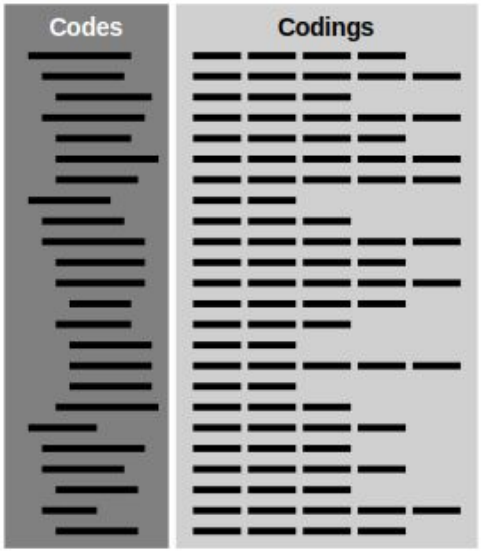
- Further coders (usually one) code the primary materials using the code system

The intercoder agreement is the overlap/agreement of the coding sets

Measures of Intercoder Agreement

DR

Codes and codings (original coder)



Codes and codings (second coder)

Measure of coder agreement

- Cohen's Kappa (κ)
- Krippendorff's Alpha (α)
- Scott's Pi
- Fleiss' Kappa

6. Iterative Process

Theory Building is Iterative

Theory building proceeds in iterations

- Iterations incrementally answer the primary research question at hand
- Triangulation and expansion improve rigor and increase relevance

Research Project Iteration

A research project proceeds in iterations

- Each iteration provides an increment
- Stopping criterion is theory saturation
- Each iteration gathers new data based on insights
- New data (samples) follow sampling model and strategy

The research project determines the primary methodology

Build-out Using Multiple Research Projects

Theory building proceeds incrementally, across many research projects

- Larger research projects are composed of smaller ones
- Multiple research projects allow for triangulation (rigor)
- Multiple research projects expand scope (relevance)

Each research project may use a different methodology

- Systematic review
- Qualitative survey
- Action research
- Case study research

7. Theory Presentation

Theory Presentation

- Prose-based descriptions
- Mathematical models / equations
- Structural equation models [1]
- Concept matrices [2]
- Handbooks [3]

[1] See Palvia et al. (2006): Research models.
[2] See Webster & Watson (2002): Analyzing the past to prepare for the future.
[3] See Riehle et al. (2025): Pattern discovery and validation (a.k.a. the handbook method).

Summary

1. QDA in theory building
2. QDA artifacts
3. QDA methods
4. Straussian coding
5. Quality assurance
6. Iterative process
7. Theory presentation

Thank you! Any questions?

dirk.riehle@fau.de – https://oss.cs.fau.de

dirk@riehle.org – https://dirkriehle.com – @dirkriehle

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